Roll No. | | | | | | | |

Odd End Semester Examination December-2025

Name of the Course: B.Tech. Biotechnology — Semester: III
Name of the Paper: Genetics and Molecular Biology — Paper Code: TBT-301

Time: 3 Hours — Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Define Mendel's law of segregation and give one suitable example.	CO 1
(b)	Write a short note on any two: a) Incomplete dominance; b) Population Genetics; c) Sex linked inheritance.	
(c)	What do you understand by Chromosomal disorders? Explain the genetic cause and features of Down syndrome.	
Q2	(20 marks)	
(a)	Mention key experiments proving DNA as the genetic material.	CO 2
(b)	Illustrate structural differences between DNA and RNA with diagrams.	
(c)	Write short notes on any two of the followings- a) Repetitive DNA; b) Transposons; c) DNA renaturation	
Q3	(20 marks)	
(a)	Describe the process of initiation of DNA replication in eukaryotes along with a labelled diagram.	CO 3
(b)	Write short notes on any two of the followings: a) Prokaryotic DNA Polymerases; b) Telomerases; c) Types of DNA damages	
(c)	How is pyrimidine dimer formed? Describe the process of repair of in case of such dimers.	
Q4	(20 marks)	
(a)	Discuss post-transcriptional modifications and their functional importance.	CO 4
(b)	How mRNA is synthesized in prokaryotes. Also give a diagram to demonstrate the process.	
(c)	Write short notes on any two of the followings: a) RNA Polymerases; b) Promoters; c) Splicing	
Q5	(20 marks)	
(a)	Discuss the role of ribonucleoprotein in prokaryotic translation.	CO 5
(b)	Write short notes on any two of the followings: a) Lac operon; b) Genetic code; c) Wobble hypothesis	
(c)	What are post translational modifications? How these modifications help proteins?	